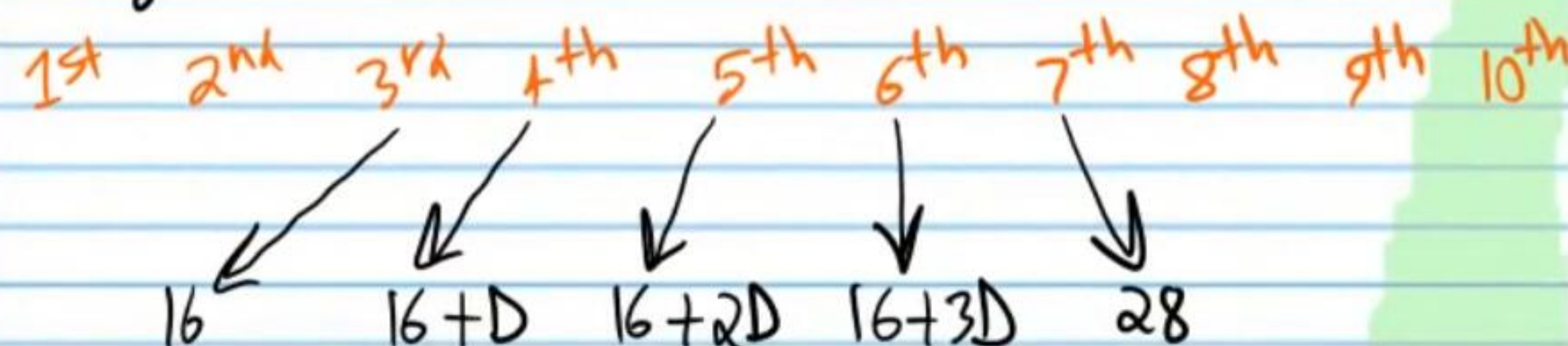


Sequences and Series (Part II)

More Problems on Arithmetic and Geometric Sequences

- ① If the 3rd term of an arithmetic sequence is 16, and the 7th term is 28, what is the 10th term?

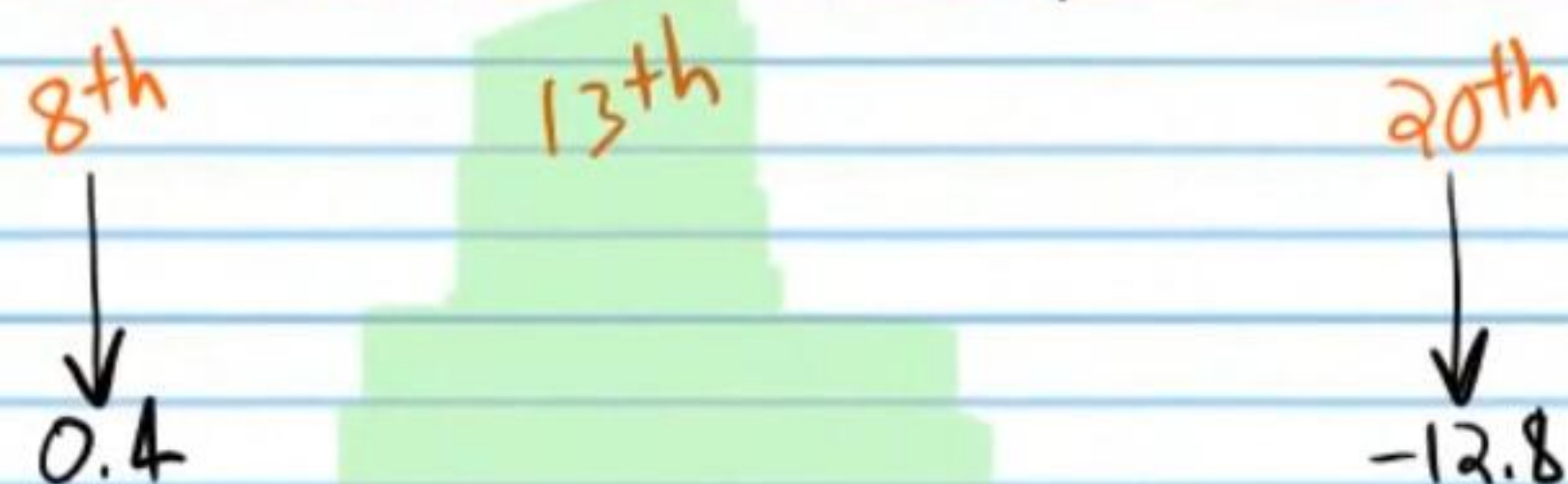


7th 8th 9th 10th

$28 + 3 + 3 + 3 = \boxed{37}$

$16 + 4D = 28$
 $D = 3$

- ② If the 8th term of an arithmetic sequence is 0.4, and the 20th term is -12.8, what is the 13th term?

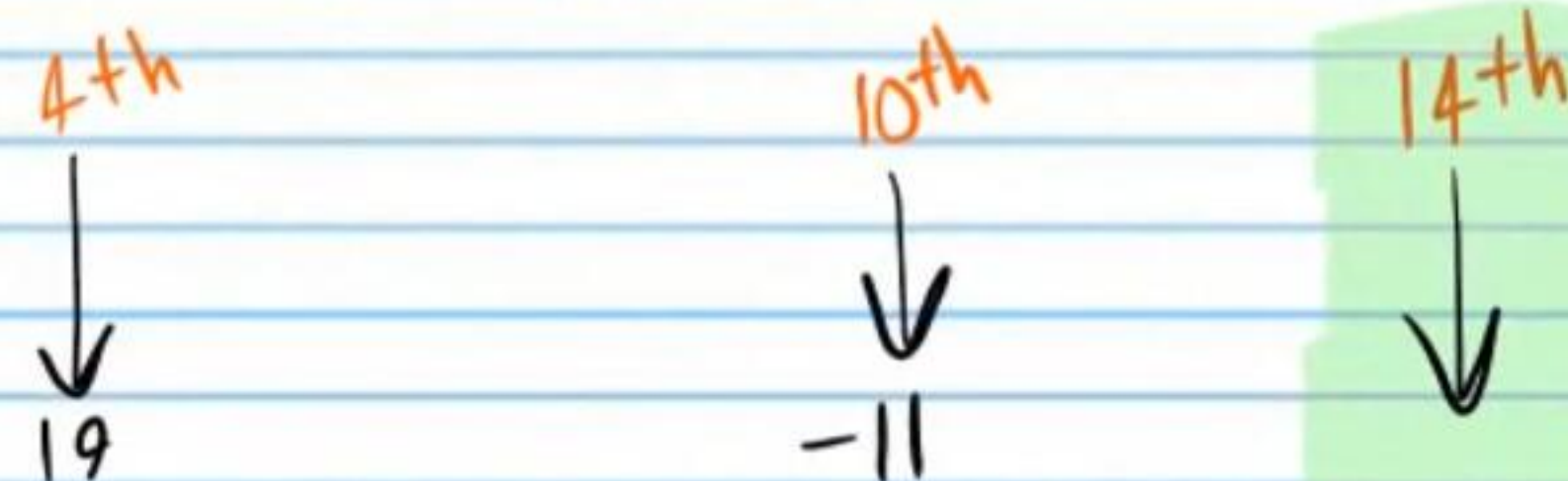


$0.4 + 12D = -12.8$
 $D = -1.1$

8th 13th

$0.4 - 1.1 - 1.1 - 1.1 - 1.1 - 1.1 = \boxed{-5.1}$

- ③ If the 4th term of an arithmetic sequence is 19, and the 10th term is -11, what is the 14th term?

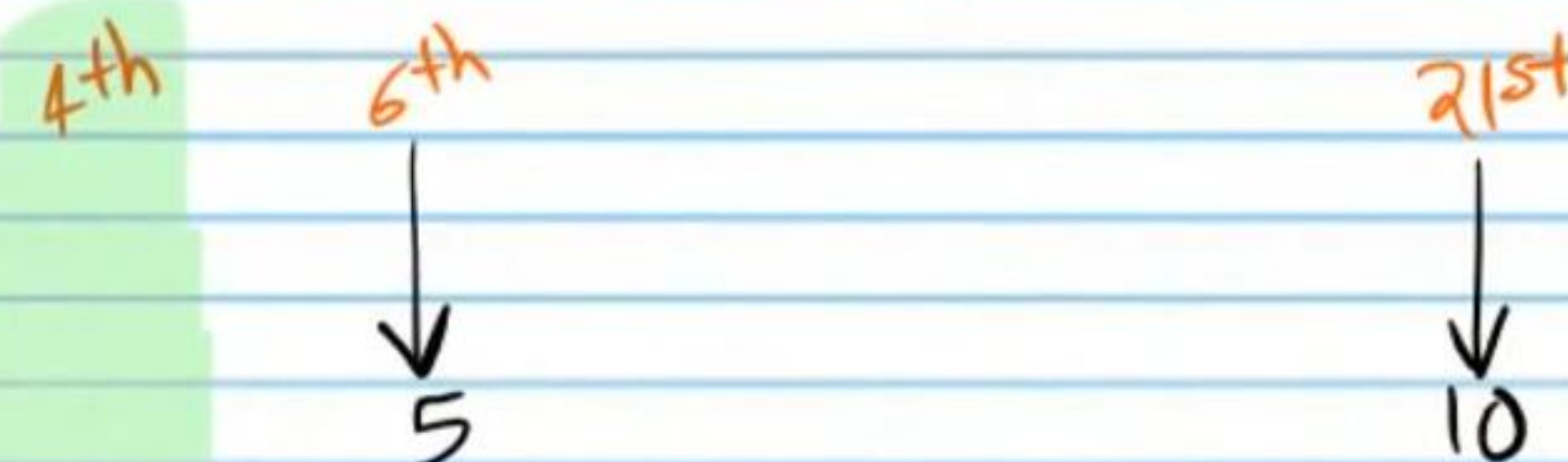


$19 + 6D = -11$
 $D = -5$

10th 14th

$-11 - 5 - 5 - 5 - 5 = \boxed{-31}$

- ④ If the 6th term of an arithmetic sequence is 5, and the 21st term is 10, what is the 4th term?



$5 + 15D = 10$
 $D = \frac{1}{3}$

4th 5th 6th

$4\frac{1}{3} = (-\frac{1}{3}) + (-\frac{1}{3}) + 5$

- ⑤ If the 5th term of a geometric sequence is 6, and the 7th term is 54, what is the 8th term? Assume that the common ratio is positive.

$$\begin{array}{cccc}
 \text{5th} & \text{6th} & \text{7th} & \text{8th} \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 6 & 6r & 54 & \\
 & & 6r^2 & \\
 & & 6r^2 = 54 & \\
 & & r^2 = 9 & \\
 & & r = 3 &
 \end{array}$$

$$\begin{array}{cc}
 \text{7th} & \text{8th} \\
 \downarrow & \downarrow \\
 54 & 54(3) = \boxed{162}
 \end{array}$$

- ⑥ If the 23rd term of a geometric sequence is $\frac{1}{9}$, and the 26th term is 24, what is the 20th term?

$$\begin{array}{cccc}
 \text{20th} & \text{23rd} & \text{24th} & \text{25th} & \text{26th} \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 & \frac{1}{9} & \frac{1}{9}r & \frac{1}{9}r^2 & 24 \\
 & & & \frac{1}{9}r^3 & \\
 & & & \frac{1}{9}r^3 = 24 & \\
 & & & r^3 = 216 & \\
 & & & r = 6 &
 \end{array}$$

$$\begin{array}{cccc}
 \text{20th} & \text{21st} & \text{22nd} & \text{23rd} \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 \frac{1}{9} \div 6^3 & \frac{1}{9} \div 6^2 & \frac{1}{9} \div 6 & \frac{1}{9} \\
 \boxed{\frac{1}{1,944}} & & &
 \end{array}$$

- ⑦ If the 6th term of a geometric sequence is -4, and the 10th term is -64, what is the 9th term? Assume that the common ratio is positive.

$$\begin{array}{ccc}
 \text{6th} & \text{9th} & \text{10th} \\
 \downarrow & \downarrow & \downarrow \\
 -4 & & -64 \\
 & & -4r^4 = -64 \\
 & & r^4 = 16 \\
 & & r = 2
 \end{array}$$

$$\begin{array}{cc}
 \text{9th} & \text{10th} \\
 \downarrow & \downarrow \\
 \boxed{-32} = -64 \div 2 & -64
 \end{array}$$

- ⑧ If the 2nd term of a geometric sequence is 27, and the 8th term is $\frac{1}{27}$, what is the 10th term? Assume that the common ratio is positive.

$$\begin{array}{ccc}
 \text{2nd} & \text{8th} & \text{10th} \\
 \downarrow & \downarrow & \downarrow \\
 27 & \frac{1}{27} & \\
 & 27r^6 = \frac{1}{27} & \\
 & r^6 = \frac{1}{729} & \\
 & r = \frac{1}{3} &
 \end{array}$$

$$\begin{array}{ccc}
 \text{8th} & \text{9th} & \text{10th} \\
 \downarrow & \downarrow & \downarrow \\
 \frac{1}{27} & \frac{1}{27} \cdot \frac{1}{3} & \frac{1}{27} \cdot \frac{1}{3} \cdot \frac{1}{3} \\
 & & \boxed{\frac{1}{243}}
 \end{array}$$

Infinite Geometric Series

Finding the Value of an Infinite Geometric Series

If r is the common ratio of an infinite geometric series, and $-1 < r < 1$, then

$$S_{\infty} = \frac{a_1}{1-r}$$

where S_{∞} is the sum of ALL of the terms in the series, and a_1 is the first term.

Proof: $S_n = a_1 \frac{1-r^n}{1-r} = \frac{a_1(1-r^n)}{1-r} = \frac{a_1 - a_1 r^n}{1-r} = \frac{a_1}{1-r} - \frac{a_1 r^n}{1-r}$

If $-1 < r < 1$, and n becomes very large, then r^n approaches zero; thus,

$$S_{\infty} = \frac{a_1}{1-r} - \frac{a_1(0)}{1-r} = \frac{a_1}{1-r} - 0 = \frac{a_1}{1-r} \quad \checkmark$$

Find the sum of all of the terms in each geometric sequence.

• ⑨ $5, \frac{1}{2}, \frac{1}{20}, \dots$

$$5r = \frac{1}{2}$$

$$r = \frac{1}{10}$$

$$-1 < \frac{1}{10} < 1 \quad \checkmark$$

$$S_{\infty} = \frac{a_1}{1-r}$$

$$S_{\infty} = \frac{5}{1 - \frac{1}{10}}$$

$$S_{\infty} = \frac{5}{\frac{9}{10}}$$

$$S_{\infty} = 5 \div \frac{9}{10}$$

$$S_{\infty} = \boxed{\frac{50}{9}}$$

⑩ $-1, \frac{4}{5}, -\frac{16}{25}, \dots$

$$-1 \cdot r = \frac{4}{5}$$

$$r = -\frac{4}{5}$$

$$-1 < -\frac{4}{5} < 1$$



$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{-1}{1 - (-\frac{4}{5})}$$

$$= \frac{-1}{1 + \frac{4}{5}}$$

$$= \frac{-1}{\frac{9}{5}}$$

$$= \boxed{-\frac{5}{9}}$$

* ⑪ $0.3, 0.33, 0.363, \dots$

$$0.3r = 0.33$$

$$r = 1.1$$

Because 1.1 is not between -1 and 1, the sum does not exist.

⑫ $-0.5, -0.35, -0.245, \dots$ • ⑬ $10, 2, \frac{2}{5}, \dots$

$$-0.5r = -0.35$$

$$r = 0.7$$

$$-1 < 0.7 < 1$$

$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{-0.5}{1-0.7}$$

$$= \frac{-0.5}{0.3}$$

$$= \boxed{-\frac{5}{3}}$$

$$10r = 2$$

$$r = \frac{1}{5}$$

$$-1 < \frac{1}{5} < 1$$

$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{10}{1-\frac{1}{5}}$$

$$= \frac{10}{\frac{4}{5}}$$

$$= \frac{50}{4}$$

$$= \frac{25}{2}$$

$$= \boxed{12.5}$$

⑭ $-2, -\frac{3}{2}, -\frac{9}{8}, \dots$

$$-2r = -\frac{3}{2}$$

$$r = \frac{3}{4}$$

$$-1 < \frac{3}{4} < 1$$

$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{-2}{1-\frac{3}{4}}$$

$$= \frac{-2}{\frac{1}{4}}$$

$$= \boxed{-8}$$

★ ⑮ $4, -20, 100, \dots$

$$4r = -20$$

$$r = -5$$

Because -5 is not between -1 and 1 , the sum does not exist.

$$\textcircled{16} -0.5, 0.125, -0.03125, \dots$$

$$-0.5r = 0.125$$

$$r = -0.25$$

$$-1 < -0.25 < 1$$



$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{-0.5}{1-(-0.25)}$$

$$= \frac{-0.5}{1.25}$$

$$= \boxed{-0.4}$$

Expressions for the n^{th} Terms of Sequences

Write the first three terms of each sequence written below. Assume that the first term is produced when $n=1$.

$$\bullet \textcircled{17} \{n^2-1\}$$

$$1^2-1, 2^2-1, 3^2-1$$

$$\boxed{0, 3, 8}$$

$$\textcircled{18} \{\cos n\} \quad n \text{ is in radian units.}$$

$$\boxed{\cos 1, \cos 2, \cos 3}$$

$$\textcircled{19} \left\{ \frac{1}{2n} \right\}$$

$$\frac{1}{2(1)}, \frac{1}{2(2)}, \frac{1}{2(3)}$$

$$\boxed{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}}$$

$$\bullet \textcircled{21} \{5n-1\}$$

$$5(1)-1, 5(2)-1, 5(3)-1$$

$$\boxed{4, 9, 14}$$

$$\textcircled{23} \{\sin^2(n)\}$$

$$\boxed{\sin^2(1), \sin^2(2), \sin^2(3)}$$

$$\textcircled{20} \{3^n + n^3\}$$

$$3^1+1^3, 3^2+2^3, 3^3+3^3$$

$$3+1, 9+8, 27+27$$

$$\boxed{4, 17, 54}$$

$$\textcircled{22} \{\sqrt{n}\}$$

$$\sqrt{1}, \sqrt{2}, \sqrt{3}$$

$$\boxed{1, \sqrt{2}, \sqrt{3}}$$

$$\textcircled{24} \left\{ \frac{4}{n+2} + \sqrt{2^n} \right\}$$

$$\frac{4}{1+2} + \sqrt{2^1}, \frac{4}{2+2} + \sqrt{2^2}, \frac{4}{3+2} + \sqrt{2^3}$$

$$\frac{4}{3} + \sqrt{2}, \frac{4}{4} + \sqrt{4}, \frac{4}{5} + \sqrt{8}$$

$$\frac{4}{3} + \sqrt{2}, 1+2, \frac{4}{5} + 2\sqrt{2}$$

$$\boxed{\frac{4}{3} + \sqrt{2}, 3, \frac{4}{5} + 2\sqrt{2}}$$

Determine if each sequence is arithmetic or geometric. Then find an expression for the n^{th} term.

• (25) 5, 7, 9, 11, ...

I) Arithmetic

II) $a_n = a_1 + D(n-1)$

$$a_n = 5 + 2(n-1)$$

$$a_n = 5 + 2n - 2$$

$$a_n = 3 + 2n$$

$$\boxed{\sum 3 + 2n}$$

(27) 56, 49, 42, 35, ...

I) Arithmetic

II) $a_n = a_1 + D(n-1)$

$$a_n = 56 + (-7)(n-1)$$

$$a_n = 56 - 7(n-1)$$

$$a_n = 56 - 7n + 7$$

$$a_n = 63 - 7n$$

$$\boxed{\sum 63 - 7n}$$

(26) 9, 3, 1, $\frac{1}{3}$, ...

I) Geometric

II) $a_n = a_1 r^{n-1}$

$$a_n = 9\left(\frac{1}{3}\right)^{n-1}$$

$$\boxed{\sum 9\left(\frac{1}{3}\right)^{n-1}}$$

(28) -1, 2, -4, 8, ...

I) Geometric

II) $a_n = a_1 r^{n-1}$

$$a_n = -1(-2)^{n-1}$$

$$\boxed{\sum -1(-2)^{n-1}}$$

• (29) 1, 5, 25, 125, ...

I) Geometric

II) $a_n = a_1 r^{n-1}$

$$a_n = (1)(5)^{n-1}$$

$$a_n = 5^{n-1}$$

$$\boxed{\sum 5^{n-1}}$$

(30) -6, -10, -14, -18, ...

I) Arithmetic

II) $a_n = a_1 + D(n-1)$

$$a_n = -6 + (-4)(n-1)$$

$$a_n = -6 - 4(n-1)$$

$$a_n = -6 - 4n + 4$$

$$a_n = -2 - 4n$$

$$\boxed{\sum -2 - 4n}$$

(31) 7.4, 8.61, 9.82, 11.03, ...

I) Arithmetic

II) $a_n = a_1 + D(n-1)$

$$a_n = 7.4 + 1.21(n-1)$$

$$a_n = 7.4 + 1.21n - 1.21$$

$$a_n = 6.19 + 1.21n$$

$$\boxed{\sum 6.19 + 1.21n}$$

(32) 3, 6.3, 13.23, 27.783, ...

I) Geometric

II) $a_n = a_1 r^{n-1}$

$$a_n = 3(2.1)^{n-1}$$

$$\boxed{\sum 3(2.1)^{n-1}}$$

The following sequences are neither arithmetic nor geometric. Use inductive reasoning to find an expression for the n^{th} term. Assume that the first term is produced when $n=1$.

• (33) $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ (34) $\pi^2, \pi^3, \pi^4, \pi^5, \dots$

$$\left\{ \frac{1}{n} \right\}$$

$$\left\{ \pi^{n+1} \right\}$$

(35) $\sin^5(e), \sin^6(e), \sin^7(e), \sin^8(e), \dots$

$$\left\{ \sin^{n+4}(e) \right\}$$

* Hint: n^3 is a part of the expression.

(36) $-2, 5, 24, 61, \dots$ (37) $\log_2 1, \log_2 2, \log_2 3, \log_2 4, \dots$

$$\left\{ n^3 - 3 \right\}$$

$$\left\{ \log_2 n \right\}$$

(38) $e, \frac{e^2}{\sqrt{2}}, \frac{e^3}{\sqrt{3}}, \frac{e^4}{\sqrt{4}}, \dots$

$$\left\{ \frac{e^n}{\sqrt{n}} \right\}$$

(39) $\frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \dots$

$$\left\{ \frac{n+2}{n+3} \right\}$$

* Hint: n^2 is a part of the expression.

(40) $2, 5, 10, 17, \dots$

$$\left\{ n^2 + 1 \right\}$$

Writing the n^{th} Term in Recursive Form

Recursive Form: An equation that shows how to find a term in a sequence using the previous term. In order for the sequence to be adequately defined, one term must be given. Usually this is the first term.

Example I: $1, 3, 5, 7, \dots$

Explicit Form: $\{2n-1\}$ Recursive Form: $\{a_1=1, a_n=a_{n-1}+2\}$

Example II: $-3, -12, -48, -192, \dots$

Explicit Form: $\{-3(4)^{n-1}\}$ Recursive Form: $\{a_1=-3, a_n=4a_{n-1}\}$

Write an equation of the n^{th} term of each sequence in recursive form. Also, show the value of the first term.

• (41) $-6, -10, -14, -18, \dots$ (42) $1, 5, 25, 125, \dots$

$$\left\{ a_1 = -6, a_n = a_{n-1} - 4 \right\}$$

$$\left\{ a_1 = 1, a_n = 5a_{n-1} \right\}$$

(43) $7.4, 8.61, 9.82, 11.03, \dots$ (44) $3, 6.3, 13.23, 27.783, \dots$

$$\left\{ a_1 = 7.4, a_n = a_{n-1} + 1.21 \right\}$$

$$\left\{ a_1 = 3, a_n = 2.1a_{n-1} \right\}$$

• (45) $2, 4, 6, 8, \dots$

(46) $9, 3, 1, \frac{1}{3}, \dots$

$$\left\{ a_1 = 2, a_n = a_{n-1} + 2 \right\}$$

$$\left\{ a_1 = 9, a_n = \frac{1}{3}a_{n-1} \right\}$$

④⑦ 56, 49, 42, 35, ... ④⑧ -1, 2, -4, 8, ...

$$\{a_1 = 56, a_n = a_{n-1} - 7\}$$

$$\{a_1 = -1, a_n = -2a_{n-1}\}$$